GP OS

→ features

Embedded OS → Customized OS.

→ Less featured OS.

① OS → Recompile → New OS

features less → Emb OS.

② Emb. OS

↳ from the Scratch

Raspbian OS (Customized Linux OS)
(Emb OS)

ucos — Emb OS

Why Emb. Linux instead Emb windows?

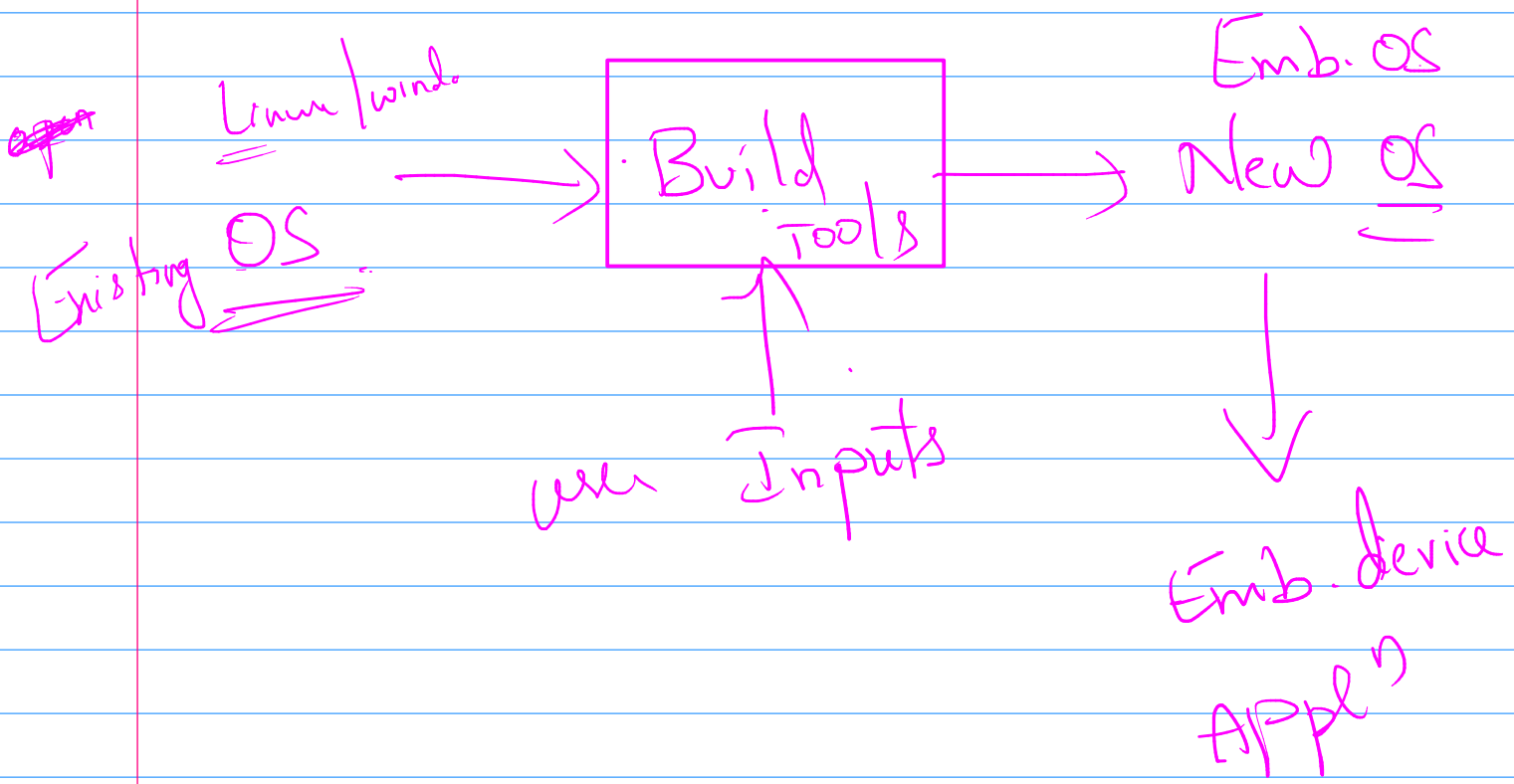
↓

Linux. open Source

↘ Customization

Buildroot / Yocto → Build Tools

Menu based Tools



Raspberry board → CPU → ARM-Cortex-A

Bluepill board / Nucleo board / Discovery board } STM32

ARM-Cortex-MX

Keil IDE | STMcube IDE

RTOS :- (Real time OS)

RTOS = Customized OS + real time Constraints

RTOS :-

VxWorks, QNX, RT Linux, RL-RTX

RL-RTX → ARM

free RTOS.

→ AVR, PIC, ARM

Keil IDE → ARM-Arch Controllers Can be programming

Keil IDE By ARM Company.

RL-RTX Library is given by Keil IDE

freeRTOS use with Keil IDE

Library should be installed.

ABS :- if (wheels locking detection)

{ Controlled

}

Video Streaming :-

1000 pixels / sec

∴ 900 / sec

Song1 = Song2